

Problem Set 5

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1 Combinatorics 1

The following problems assume a standard 52 card deck with 13 *ranks*, $R = \{A, 2, 3, 4, 5, 6, 7, 8, 9, 10, J, Q, K\}$ each of which comes in each of four *suits*, $S = \{\clubsuit, \diamondsuit, \heartsuit, \spadesuit\}$. You do not need to simplify your answers.

Problem 1: How many ways are there to choose one card from each rank (for a total of 13 cards), such that not every card is red ($\diamondsuit$ or $\heartsuit$)?

Problem 2: How many cards do I need to draw in order to guarantee that I draw a three-of-a-kind (3 or more cards of the same rank)?

Problem 3: How many ways are there for me to draw 10 cards from the deck and order them from left to right?